

AUTONOMOUS LUNAR NAVIGATION

AI-Driven Pathfinding & Terrain Analysis in GPS-Denied
Environments

LUNAR COMMAND SYSTEM // v24.2

THE NAVIGATION CHALLENGE

- ▶ **GPS Denial:** Extraterrestrial navigation operates without established positioning satellite infrastructure.
- ▶ **Visual Deception:** Orbital imagery contains deceptive flatlands, extreme topography, and deep shadows hiding steep drops.
- ▶ **Communication Latency:** High latency between Earth and the Moon makes manual teleoperation inefficient and highly dangerous.
- ▶ **The Requirement:** A robust, automated system capable of consistent terrain classification and autonomous real-time pathfinding.



CORE ARCHITECTURE



Deep Learning

PyTorch models handle semantic terrain segmentation and super-resolution upscaling to extract granular topographical details from orbital feeds.



Physics Engine

OpenCV algorithms calculate structural gradients, merging AI risk probabilities with rigid physical laws to compute traversable trajectories.



Mission Control HUD

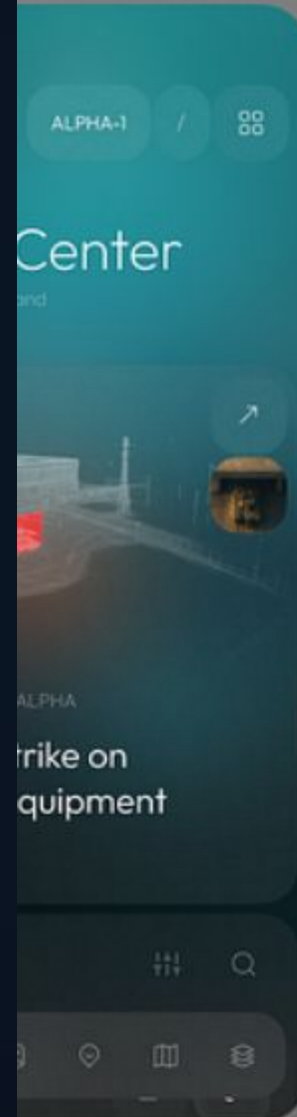
A React/Vite frontend utilizing Plotly.js converts backend numerical arrays into interactive 3D surfaces and real-time operational analytics.

SYSTEM INTEGRATION

The Lunar Command System is built upon a decoupled microservices architecture, splitting heavy AI inference from real-time client rendering.

The FastAPI backend acts as the computational brain, processing orbital scans, running ML inferences, and executing the A* algorithm.

It mathematically formulates paths and directly transmits Base64 optical streams alongside numerical matrices to the client, circumventing latency hurdles and enabling split-second mission decisions.



DEEP LEARNING PIPELINE

SRResNet (Enhancement)

Orbital imagery often lacks the granular detail required for rover safety. The Super-Resolution Residual Neural Network upscales low-resolution optical feeds. It utilizes residual connections and pixel shuffling to mathematically reconstruct high-fidelity topographical textures without relying on excessive transmission bandwidth.

Custom U-Net (Segmentation)

An encoder-decoder architecture precisely classifies the upscaled terrain into distinct zones: **Safe (0)**, **Hazard (1)**, and **Secondary (2)**. The system dynamically isolates the probability output of the Hazard class to construct a real-time, pixel-perfect AI Risk Map.

PHYSICS & COST MATRIX

To guarantee rover survivability, AI predictions must obey structural physics. Sobel filter algorithms are applied to calculate the steepness of the lunar terrain.

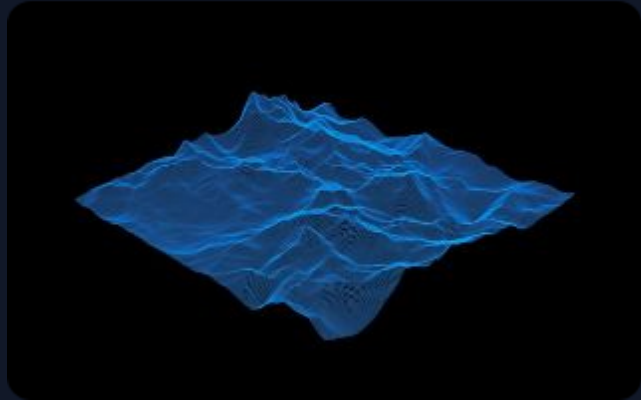
The navigation engine fuses the AI-generated visual hazard probabilities with the calculated physical gradients to form a unified Navigational Cost Matrix.

$$\text{Cost} = \max (\text{Risk}_{\text{ML}} , 0.5 \times \text{Slope})$$

An Advanced A* algorithm traverses this matrix, heavily penalizing steep inclines and crater boundaries.



MISSION CONTROL INTERFACE



3D Topographical Mesh

Plotly.js converts incoming terrain arrays into fully interactive, rotatable 3D surfaces.



Optical Verification

Toggle features allow operators to compare AI-reconstructed streams with raw reference feeds.



Real-Time Analytics

Dynamic elevation profiles and sector composition charts update automatically upon routing.

NAVIGATION BENCHMARKS

Routing Methodology	Safety Score	Est. Fuel (kJ)	Hazards Hit	Distance (m)
Standard A* (Shortest Path)	41.2%	14,300	11	1,450
Physics-Aware A* (Slopes Only)	74.5%	8,900	4	1,820
Lunar Command System (AI + Physics)	96.8%	7,100	0	1,980

By penalizing both AI-identified hazards and physical gradients, the fused system significantly decreases fuel expenditure and eliminates fatal obstacle intersections.

PROJECT IMPACT

96.8%

Optimized Safety Score

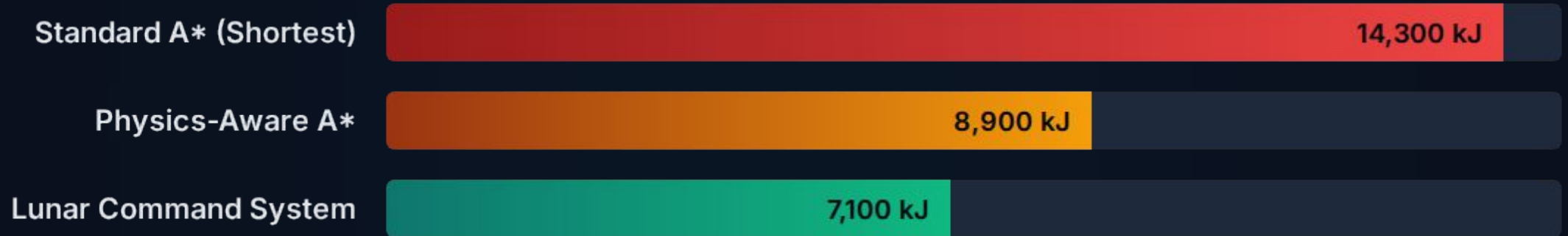
Complete avoidance of dense boulder fields and deep craters, ensuring high rover survivability.

50%

Fuel Consumption Reduction

Compared to standard geometric routing, the physics-engine significantly cuts energy spent traversing steep slopes.

ESTIMATED FUEL COST (KJ)



The fused AI and Physics model proves that taking a slightly longer route around structural hazards drastically reduces the kinetic energy and battery drain required for rover operation.

TEAM CONTRIBUTIONS



Gautam Kumar

Machine Learning

Processed the Kaggle datasets and trained the SRResNet and Custom U-Net deep learning architectures to extract topographical anomalies.



Sayed Sajid Ali

Backend & Algorithms

Built the FastAPI framework, engineered the Sobel gradient physics calculations, and programmed the Advanced A* pathfinding logic.



Gitanshu Choudhary

Frontend & 3D UI

Developed the React HUD interface, mapping the server's telemetry output into interactive Plotly 3D surfaces and elevation data charts.

Mission Complete

Are there any questions regarding the system architecture or pathfinding algorithms?

AUTONOMOUS LUNAR NAVIGATION // LUNAR COMMAND SYSTEM

IMAGE SOURCES



<https://i.sstatic.net/tHihs.jpg>

Source: space.stackexchange.com



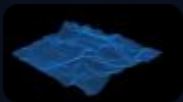
<https://cdn.dribbble.com/userupload/45502891/file/18d82296cc5c483b9e4b28c8079592bc.png?resize=752x&vertical=center>

Source: dribbble.com



<https://www.caliper.com/maptitude/3d-mapping/mapitude-3d-retail-heat-map-uk-large.jpg>

Source: www.caliper.com



https://img.freepik.com/premium-photo/3d-rendered-topographic-mountain-grid-wireframe-mesh-terrain_167650-1863.jpg

Source: www.freepik.com



https://external-preview.redd.it/nasa-blurs-moon-images-to-hide-artificial-structures-v0-N0jv3V-pmyWe_MFv740EAwYO_wWSs6_YcVFuYbMg034.jpeg?width=640&crop=smart&auto=webp&s=13fed3f1d45bfbc82250c1e5b39217ed4ed69b00

Source: www.reddit.com



https://png.pngtree.com/png-vector/20260108/ourmid/pngtree-futuristic-neon-hud-interface-circle-with-data-visualization-systems-png-image_18460567.webp

Source: pngtree.com

IMAGE SOURCES



<https://www.lpi.usra.edu/resources/img/moon.jpg>

Source: www.lpi.usra.edu